

ROBOPharma GUI

User Manual

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WRITTEN BY ROBERTA GRIFFITHS

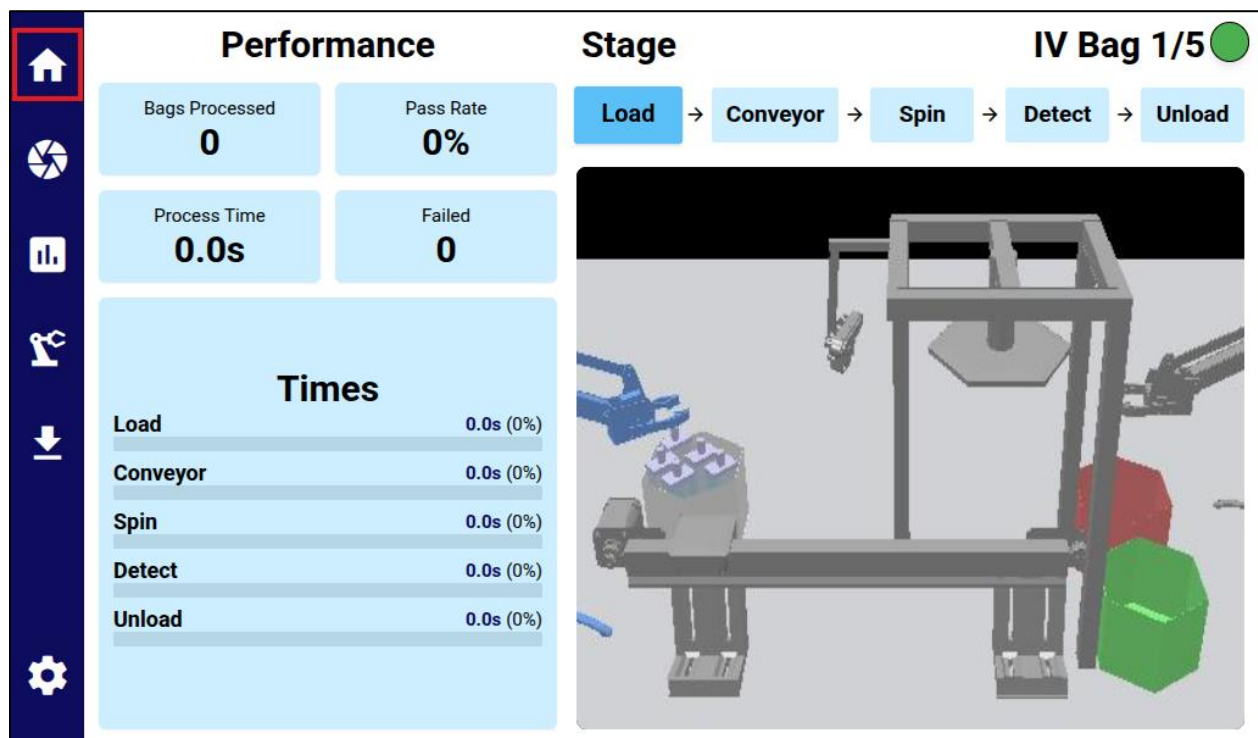
01

HOW TO LAUNCH THE GUI

The desktop app will launch all required parts of the system, the app will automatically load.

02

HOME PAGE



This is the original view the GUI will launch to, and can be accessed by touching the home icon in the navigation bar (shown in red).

Performance

- **Bags Processed:** Shows total number of bags processed over a session

- **Pass Rate:** Percentage of bags that passed (i.e. clean or non contaminated bags)
- **Process Time:** Total number of seconds passed for the current cycle (involving active stage times and idle time)
- **Failed:** Number of bags that failed (i.e. contaminated) across the time the GUI is running

Times

- Shows number of seconds taken for each stage of the process and its percentage of all the time taken for the whole cycle
- This resets every cycle

Stage

- Shows the active stage of the process, which is highlighted in a brighter blue

Simulation

- Shows the camera feed from the CoppeliaSim simulation

03

DETECTION PAGE

The screenshot displays the Detection Page interface. On the left is a navigation bar with icons for Home, Camera (highlighted in red), Bar Chart, Wrench, Download, and Settings. The main panel shows the following data:

- ID : 1** with a **FAIL** status in a red box.
- A circular progress indicator showing a small green segment and a large red segment.
- Pass: Fail:** 0 1
- Confidence:** 0.727%
- Contamination:** MEDIUM
- Particles:** 2
- Bubbles:** 5
- Analysis Time:** 6.0s

Below this is an **Information** section:

- Model:** faster_rcnn
- Particle Threshold:** 1
- Confidence Threshold:** 0.9985

On the right is a camera feed showing a bag of **SODIUM CHLORIDE Inject'n USP, 0.3%**. The bag label includes details like REF L7600, 300 mL, IDC 0265-7213-00, DIN 22614, and a barcode.

This is the page that is accessed by clicking on the camera icon in the navigation bar (highlighted in red). This displays information relating to the detection stage.

Detection Results

- The ID in the top left shows which bag the information is for
- The FAIL badge in the top right shows the status of the bag (FAIL = contaminated, PASS = clean or non contaminated)
- The pie chart updates dynamically to show how many bags have passed or failed (green for pass, red for fail)
- The number of bags passed and failed is also shown in text
- The rest of the information shows the information associated with the detection process for the current bag

Information

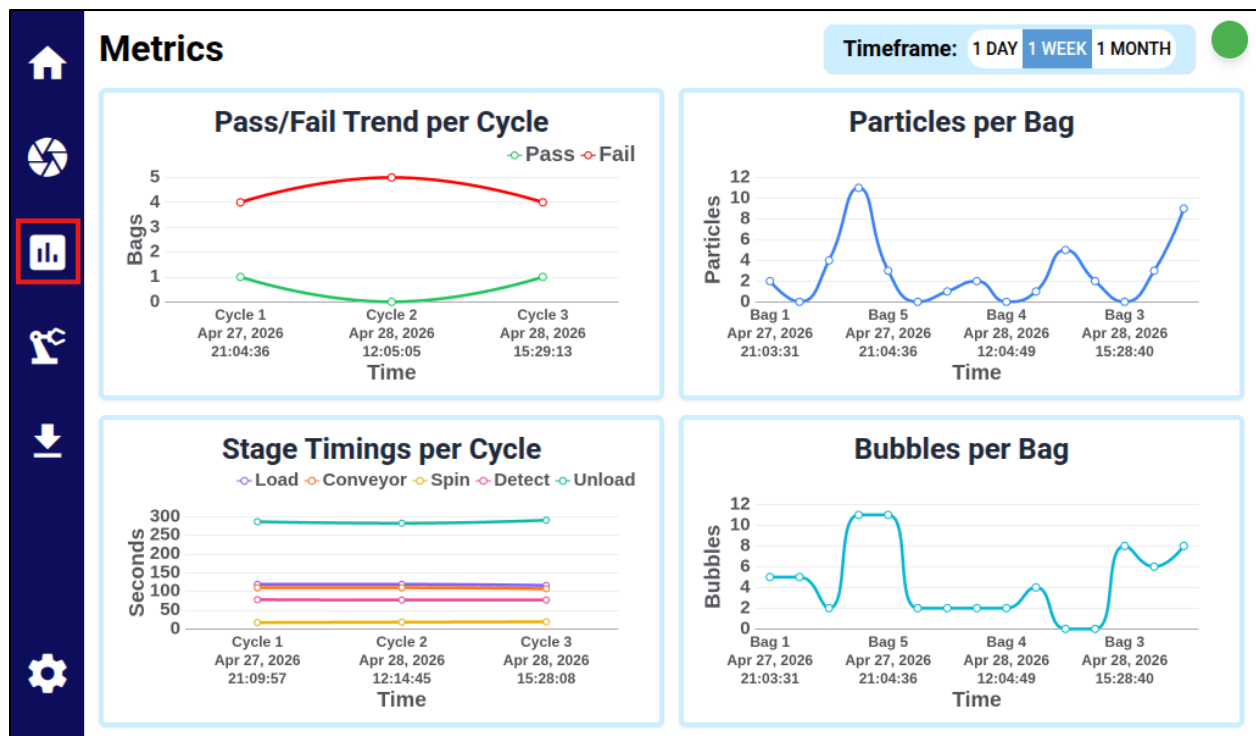
- This shows the detection setup variables, including the model user and the particle and confidence thresholds

Camera Stream

- This shows the real-time camera livestream which is used for the detection process, including the bounding boxes for particles and bubbles

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METRICS PAGE



This page shows metrics associated with the rig. It can be accessed by the graph icon in the navigation bar which is highlighted in red.

Timeframe

- Located in the top right, this controls when the time range of the data, there are options to show data from the previous day, week, or month

Pass/Fail Trend per Cycle

- For each cycle this traces the number of bags which have passed or failed
- The X axis shows the time the cycle was run at

Particles per Bag

- For each bag this shows the number of particles detected

Stage Timings per Cycle

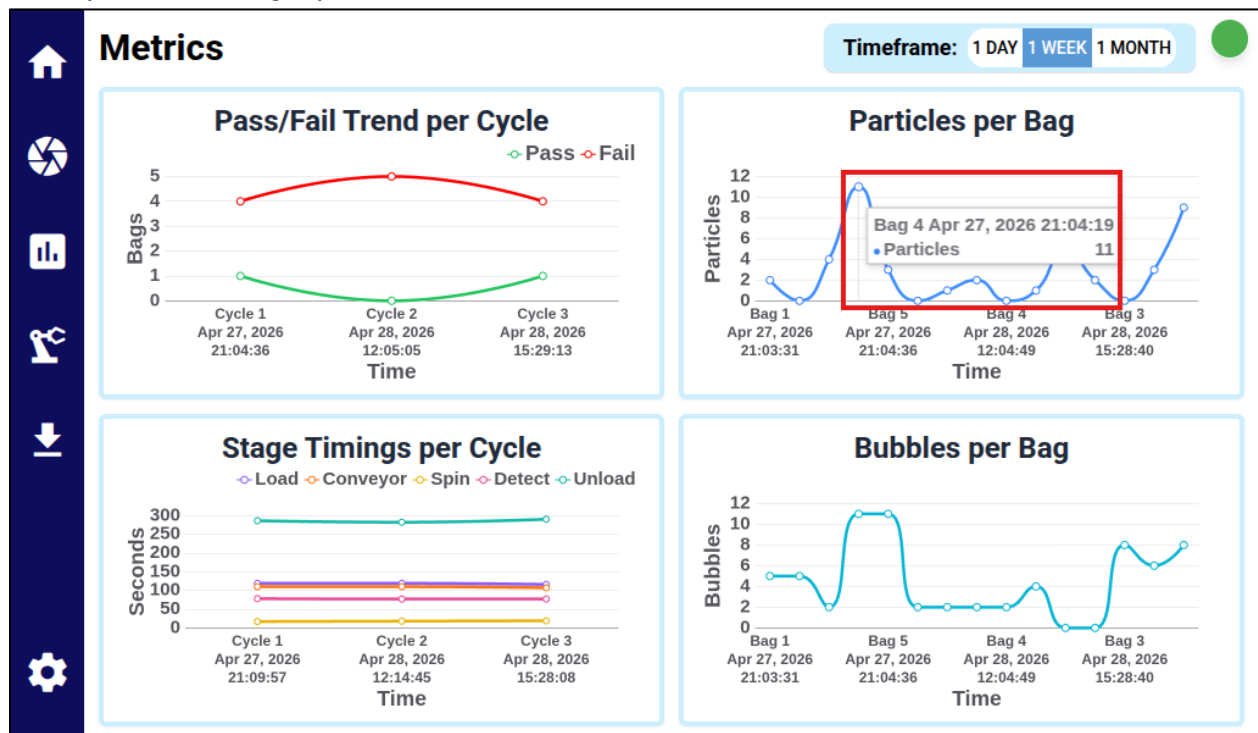
- This shows the stage timings for each cycle

Bubbles per bag

- This shows the number of bubbles detected per bag

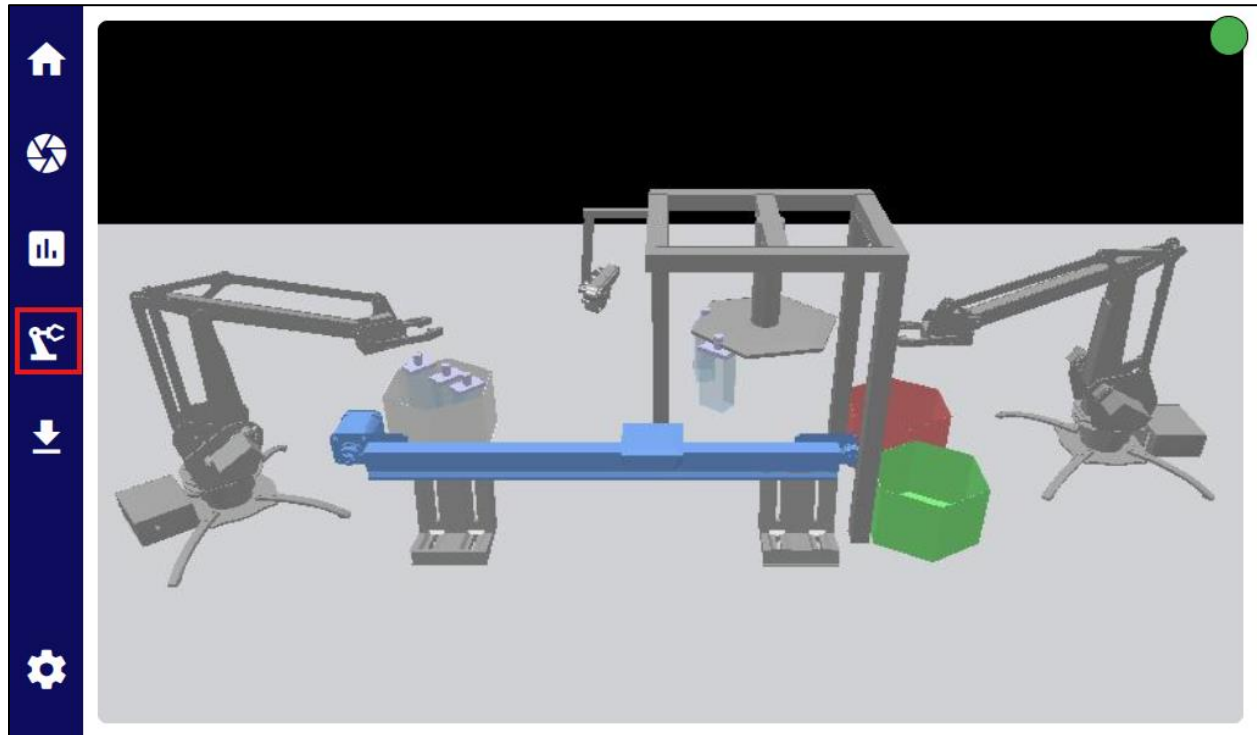
For plots that track multiple variables, items can be clicked in the legend to add or remove them from the plot.

Each point on the graph can be clicked to show further details as shown below.



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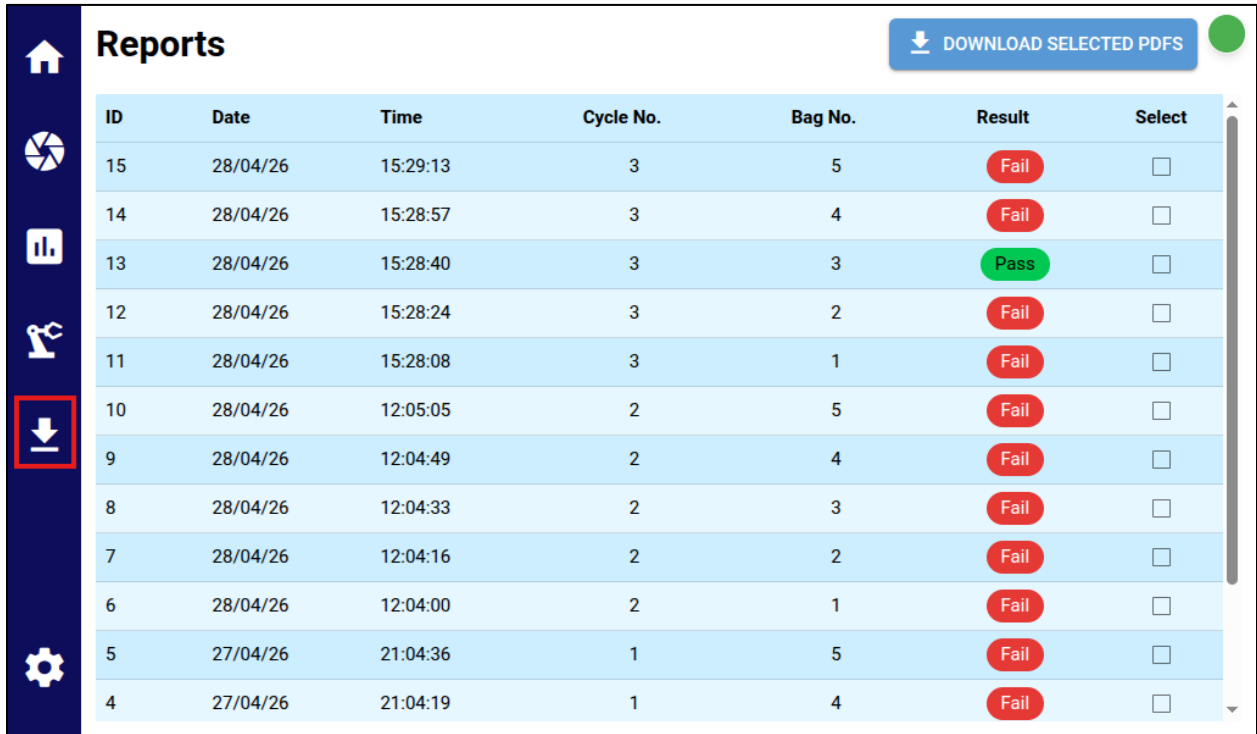
SIMULATION PAGE



This page shows a full screen view of the simulation. The active parts are highlighted using a blue colour, as seen on the conveyor above.

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DOWNLOADS PAGE



ID	Date	Time	Cycle No.	Bag No.	Result	Select
15	28/04/26	15:29:13	3	5	Fail	☐
14	28/04/26	15:28:57	3	4	Fail	☐
13	28/04/26	15:28:40	3	3	Pass	☐
12	28/04/26	15:28:24	3	2	Fail	☐
11	28/04/26	15:28:08	3	1	Fail	☐
10	28/04/26	12:05:05	2	5	Fail	☐
9	28/04/26	12:04:49	2	4	Fail	☐
8	28/04/26	12:04:33	2	3	Fail	☐
7	28/04/26	12:04:16	2	2	Fail	☐
6	28/04/26	12:04:00	2	1	Fail	☐
5	27/04/26	21:04:36	1	5	Fail	☐
4	27/04/26	21:04:19	1	4	Fail	☐

This page connects to a database of all stored records from previous cycles, accessed by the download icon highlighted in the navigation bar.

Table

- Can be sorted by clicking on the headings, which adjusts the order of items show
- The rows can be selected using the checkbox on the right side of table

Download selected PDFs

- This exports all selected rows in the table into a PDF format

Report Popup

- This is activated by clicking on a row from the table, and shown in the image below
- The images show the a picture from the camera feed from the detection, in position 1 and position 2
- The badge on the top right shows the status of the IV bag (PASS/FAIL)
- The rest of the information shows further detail of how many particles were detected and which model was used for the detection
- The report can also be exported to a PDF from this screen

Reports DOWNLOAD SELECTED PDFS

Detailed Report - Cycle 1 Bag 5

Position 1



Position 2



FAIL

Timestamp	2026-04-27 21:04:36
Particles	3
Bubbles	11
Confidence	0.50%
Contamination	MEDIUM
Model	faster_rcnn

EXPORT TO PDF

Position	Time	Status
1	27/04/26 21:03:31	Fail

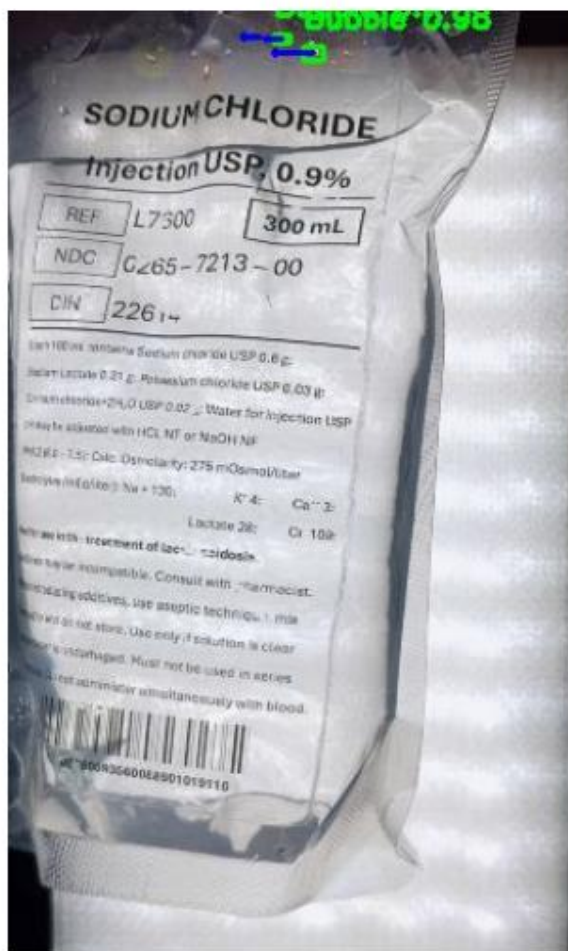
Report PDF

- The image below shows the downloaded PDF from the reports table
- This shows the same images as in the popup
- For each bag this details the information of the detection result

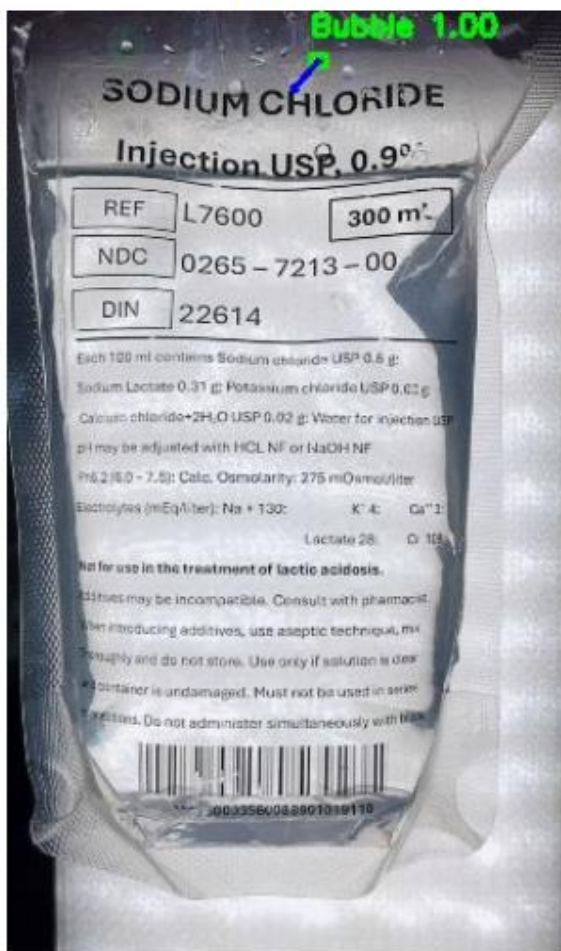
Detection Report: Cycle 1 - Bag 5

Metric	Value
Record ID	5
Timestamp	2026-04-27 21:04:36
Status	CONTAMINATED
Contamination Level	MEDIUM
Recommendation	REJECT - Particles detected
Total Particles	3
Total Bubbles	11
Confidence Score	0.50%
Detection Model	faster_rcnn

Position 1

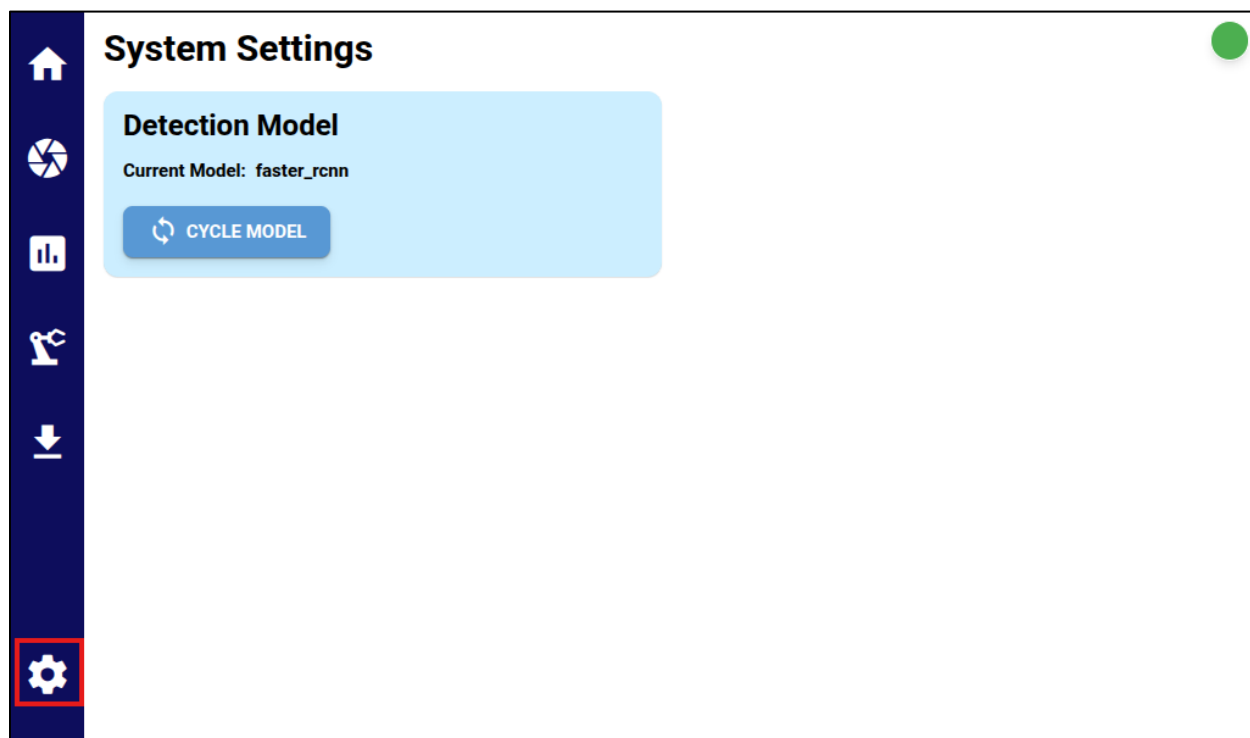


Position 2



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SETTINGS PAGE



This page is accessed from the settings cog in the navigation bar as highlighted in red.

Detection Model

- The Current Model shows the model that is currently selected to carry out the detection process
- When the CYCLE MODEL button is pressed a request is sent to change the model
- The new model name will appear in the Current Model

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ERRORS

The first image shows the popup that occurs when an error takes place. It details what the error is. If the error is not corrected the status bar will remain the error colour, as shown in the second image.

If the error is corrected (simulation is restarted) the status bar will turn green.

When an error occurs, the text popup will stay for 5s then disappear. The status indicator light can be clicked to show the text again.

Home

Camera

Bar Chart

Wrench

Download

Settings

Performance

Bags Processed

0

Pass Rate

0%

Process Time

0s

Failed

0

Times

Load	0.0s (0%)
Conveyor	0.0s (0%)
Spin	0.0s (0%)
Detect	0.0s (0%)
Unload	0.0s (0%)

ERROR: simulation stopped running

Load

→

Conveyor

→

Spin

→

Detect

→

Unload

Home

Camera

Bar Chart

Wrench

Download

Settings

Performance

Bags Processed

0

Pass Rate

0%

Process Time

0s

Failed

0

Times

Load	0.0s (0%)
Conveyor	0.0s (0%)
Spin	0.0s (0%)
Detect	0.0s (0%)
Unload	0.0s (0%)

Stage

IV Bag 0/0

Load

→

Conveyor

→

Spin

→

Detect

→

Unload